

CREDIT CARD FINANCIAL ANALYSIS

Power BI & DAX Project

1. Project Name

Credit Card Financial Analysis Dashboard

2. Project Overview

This project analyses credit card customer and transaction data using Microsoft Power BI, Power Query and DAX. The dashboard provides an interactive view of financial performance, customer behaviour, transaction activity and other key business metrics. The objective is to transform raw data into meaningful insights that can support data-driven business decision-making.

3. Business Objective

The objective of this project is to analyse credit card customer and transaction data, identify important financial trends and customer segments, understand spending behaviour, and identify factors contributing to overall financial performance. The analysis is designed to support business decisions through measurable insights and recommendations.

4. Business Questions

- What are the key revenue and transaction trends?
- Which customer segments contribute the highest revenue?
- Which customer groups have the highest transaction activity?
- Which spending categories contribute most to overall revenue?
- How does customer behaviour vary across demographic segments?
- Which areas show stronger or weaker financial performance?
- What changes in customer or transaction behaviour should receive business attention?
- What business actions can be considered based on the findings?

5. Dataset

The project uses credit card customer and transaction data containing information required to analyse customer characteristics, transactions, spending behaviour and financial performance. The dataset was prepared and transformed before being used for Power BI analysis.

Dataset source: Enter the actual dataset source used in the project.

Main tables used: Enter the actual table names from your Power BI model.

Important fields: Enter the main customer, transaction, date and financial fields used.

6. Tools and Technologies

- Microsoft Power BI – dashboard development, interactive reporting and visualization
- Power Query – data cleaning and transformation
- DAX – measures, calculations and KPI analysis

7. Data Preparation

The data was prepared before analysis to improve consistency and usability. The preparation process included checking data types, reviewing data quality, handling relevant missing or invalid values where applicable, standardising fields and creating the required structure for analysis.

Actual preparation steps performed in this project:

- Reviewed the Customer and Credit Card tables and verified the required columns and data types.
- Checked the data for missing or invalid values relevant to the analysis.
- Created the Age Group field to categorize customers into age-based segments.
- Created the Income Group field to classify customers based on income levels.
- Created the Week_Num field to support weekly transaction and revenue analysis.
- Created the required calculated/derived fields used in the dashboard analysis.
- Standardized the fields required for customer, transaction, revenue, and demographic analysis.
- Prepared the data model by connecting the Customer and Credit Card tables using Client_Num

8. Data Model

The Power BI data model consists of two main tables:

1. Customer Table

The Customer table contains customer-level information such as age group, gender, customer age, education level, occupation, car ownership, dependent count and customer satisfaction score.

2. Credit Card Table

The Credit Card table contains credit-card-related information such as card category, credit limit, annual fees, activation status, average utilization ratio, customer acquisition cost and delinquency information.

The two tables are connected using the common field **Client_Num**, which enables customer attributes to be analysed together with credit-card and financial information.

9. Table Relationship

The relationship between the tables is defined as follows:

From Table	Column	To Table	Column	Cardinality	Cross-Filter
Credit Card	Client_Num	Customer	Client_Num	One-to-One (1:1)	Both

The relationship is active and uses Client_Num as the common key. The bidirectional filtering allows selections made using customer attributes to interact with the credit-card analysis and vice versa.

10. DAX Measures and KPIs

DAX measures were created to calculate key financial and customer metrics used in the dashboard. The measures help convert raw transaction data into meaningful business indicators.

Total Revenue =

SUM(CreditCard[Annual_Fees]) +

SUM(CreditCard[Total_Trans_Amt]) +

SUM(CreditCard[Interest_Earned])

Purpose:

The Total Revenue measure calculates the overall revenue generated from annual fees, transaction amounts and interest earned.

Business Meaning:

This KPI provides an overall view of the financial contribution represented in the dataset. It can be used to compare revenue across customer segments, card categories, age groups and income groups.

Dashboard Usage:

The measure is used in the dashboard to analyse total revenue and its distribution across different customer and credit-card segments.

11. Analytical Methodology

The analysis followed an end-to-end workflow: data preparation, data modelling, KPI creation, exploratory analysis, customer analysis, transaction analysis, financial trend analysis and business interpretation. The analysis focuses on identifying measurable patterns and translating those patterns into business insights.

12. Dashboard and Output

The Power BI dashboard provides interactive views of key financial and customer metrics. Filters and visualisations allow users to explore performance across relevant customer, transaction, demographic, category and time dimensions.

14. Key Findings

The key findings should be supported by actual numbers from the dataset and dashboard. The following statements should be replaced with the calculated values from the project.

- **Revenue:** Total revenue generated is 56.52M, with revenue distributed across different age groups, income groups, and credit-card categories.

- **Transactions:** Total transaction amount is approximately 46M, indicating a significant volume of credit-card transactions during the analysed period.
- **Customers:** The dashboard contains approximately 10.293K customers.
- **Customer Segments:** The High-Income group contributes the highest revenue among the income groups, followed by the Medium and Low-income groups.
- **Spending Categories / Card Categories:** The Blue card category contributes the largest share of revenue at approximately 83.49% (47.19M), followed by the other card categories.
- **Demographics:** The 46–55 age group contributes the highest revenue, followed by the 36–45 age group. The younger age groups contribute comparatively less revenue.

15. Business Impact

The findings help the business understand which customer groups, transaction patterns and financial areas contribute most to performance. Quantifying these patterns can support more focused customer engagement, performance monitoring and business planning.

16. Business Recommendations

Based on the findings, the business can consider targeted actions around high-value customer segments, customer engagement, transaction performance and categories that materially influence financial results.

- **Focus on high-value customers:** Develop targeted offers and loyalty programs for customers in the **High-Income** group, which contributes the highest revenue.
- **Strengthen the Blue card category:** Since the Blue card category contributes approximately 83.49% of total card-category revenue, the business can focus on customer retention, usage, and relevant offers for this category.
- **Target high-revenue age groups:** The **46–55 age group** generates the highest revenue, followed by the **36–45 age group**. Targeted products and campaigns can be designed around the preferences and spending behaviour of these customer groups.
- **Improve engagement with lower-contributing segments:** Analyse the needs and transaction behaviour of lower-revenue income and age groups and use targeted campaigns to improve their card usage.
- **Monitor revenue and transaction performance:** Continue tracking total revenue, transaction amounts, card categories, income groups, and age groups through the Power BI dashboard to support regular performance monitoring and business planning.

17. Technical Highlights

- Built an interactive Power BI dashboard for credit card financial analysis.
- Used Power Query for data preparation and transformation.
- Used DAX measures to calculate relevant financial and customer KPIs.
- Created and used a Power BI data model and table relationships.
- Analysed customer, transaction, demographic and financial dimensions.

- Translated analytical results into business findings and recommendations.

18. Limitations

The analysis is based on the available dataset, its fields and the time period represented in the data. The findings may not capture factors that are not included in the dataset, such as external economic conditions, customer-specific circumstances or other business factors. Therefore, the insights should be interpreted within the scope of the available data.

19. Conclusion

This project demonstrates the use of Power BI, Power Query and DAX to transform credit card customer and transaction data into an interactive analytical dashboard. The analysis combines technical data preparation and modelling with KPI analysis, customer and transaction insights, quantified findings and business recommendations. The project demonstrates an end-to-end approach to using data to understand business performance and support decision-making.

20. GitHub Repository

GitHub Repository: https://github.com/Mahanand4/Credit_Card_Financial_Analysis_PowerBI

The repository contains the project documentation, Power BI dashboard file, final project PDF and supporting project files.

21.Submitted By

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